Experiment: Hooke's Law (Finding Spring Constant k)

Theory :

Hooke's Law relates the extension of a material to the load it is under. In other words the extension of a material beyond its equilibrium length is directly proportional to the force acting on it.

Force = spring constant x extension, **$F = k \cdot \Delta x$**

Procedure

Find the spring constant of the spring in the experiment by gathering results using the measuring tools provided. You may use the table below to record them. Then use the MS excel or a graph paper(provided) to plot your results. From the gradient(slope) you can find the spring constant

PLEASE DO NOT EXCEED THE ELASTIC LIMIT OF THE SPRING.

Results

Spring 1

Starting scale reading _____cm

Spring 2

Starting scale reading _____cm

Mass (kg)	Weight (N)	Stretch in Δ x (cm)	Stretch Δ x (m)		Mass (kg)	Weight (N)	Stretch in Δ x (cm)	Stretch Δ x (m)
0.05					0.05			
0.07					0.07			
0.09					0.09			
0.11					0.11			
0.13					0.13			

Questions:

1. What is the spring constant (k) is for each spring? What are the units of k? Which spring is stiffer?

2. Why are the cables of a suspension bridge elastic?

3. Hooke's Law is a direct relationship. What does this mean?